Presentation 2

Link to presentation PDF, and code:

Link to presentation

Code can be found in presentation2 github folder

What did each of the team members do?

- *Coders: Alyssa*
- *Analyst: Lily and Bella*
- *Data Engineer: Callie Berthold*
- *Consultant: Lauren*

Issues faced that haven't been resolved:

- Bella: Text crowding for Top 5 Patches per slice Images
- Callie: Flipped patches didn't reduce memorization. Both models still drove training loss to about 0.

Proposed next steps (not mandatory for presentation 1)**:**

- Bella: I tried changing the font size, thickness, and text placement, but all seemed to still have overlap. It may be more useful to have a legend for patches instead, that holds the information for specific attention weight. It depends on the visual that the stakeholders would want. Also, currently the script runs manually, with specific file paths for the .tif files typed out. For next steps, I would run the script automatically within attention analysis, so every test slice gets a figure. I would also consider combining this code with the ideas for the Analyst Choice 2 task.
- Callie: Try bag-level augmentation: train on a random 50–70% of each slice's patches each epoch, so the model can't memorize whole cases. I also recommend incorporating the missing patches to make the 93-case splits, so add the new cases and rerun the comparison.
- Lauren: I'd recommend embedding every patch with frozen UNI2 two ways, once resized to 224 px and once at native size padded to a multiple of 14, and training both on the same 5 splits with KimiaNet as the baseline. We should compare accuracy, benign and high-grade recall, and AUC per split, and pick whichever approach wins across the splits. I'd also recommend timing the native-size

approach, since our largest patches need about 84× more squares than resized ones.

References

Whose work did you build up on?

Lily: Built on top of starter code presentation 2

Bella: Built on top of starter code presentation 2

Alyssa: Built on Presentation 2 basecode

Callie: Build up on existing patch slices, presentation 2 base code